



PSDL Project Report

Academic Year 2024-25

Smart Face Attendance System

UCE202301 - Adwita Jain

UCE2023602 - Harshali Agarwal

UCE2023603 - Ahilya Kerle

UCE2023608 - Nidhi Borkar

Synopsis

1. Group Member Roll Numbers, CNumbers and Name:

UCE2023601 - Adwita Jain

UCE2023602 - Harshali Agarwal

UCE2023603 - Ahilya Kerle

UCE2023608 - Nidhi Borkar

2. Problem Statement:

Develop a real-time face recognition based attendance system with basic anti-spoofing and a user-friendly web interface to automatically log student attendance for a university department. The system should allow **face registration**, initiate **attendance taking via webcam**, and **store attendance records** for later viewing.

3. Keywords:

Face Recognition, Real-time Attendance, Webcam, Anti-Spoofing, Web Interface, Attendance Logging, Computer Vision, Face Recognition, Attendance System, OpenCV, LBPH Algorithm, SQLite, Tkinter, Python, Image Processing, Data Visualization

4. Abstract:

The project aims to develop a **Face Recognition-based Attendance System** using Python. It captures and recognizes student faces in **real-time**, marking attendance automatically. The system uses OpenCV's **LBPH algorithm** for face recognition, stores attendance data in an **SQLite database**, and provides **data visualization** through graphs using matplotlib and pandas. This project ensures **accuracy**, **reduces chances of proxies**, and **saves time** in institutional environments. Upon successful and verified recognition, the system automatically logs the student's attendance, including their **name, date, and time**. A user-friendly web interface, provides functionalities for new user face registration, initiating the attendance process, and viewing the recorded attendance logs. This system offers a streamlined and secure alternative to traditional manual attendance methods.

5. Module-wise Description:

- **tkinter (tk):** This module is used for creating the Graphical User Interface (GUI) of the application. It provides widgets like buttons, labels, and entry fields for user interaction.
- **tkinter.messagebox:** This sub-module of tkinter is used to display pop-up message boxes for displaying information, warnings, and errors to the user.
- **cv2 (OpenCV):** This module is used for computer vision tasks such as image and video capture, face detection, image processing, and face recognition.
- **sqlite3:** This module is used for interacting with an SQLite database to store and retrieve attendance data.
- **datetime:** This module is used for working with date and time information, specifically for recording the date and time of attendance.
- **threading:** This module is used to create and manage threads, allowing the face recognition process to run concurrently without freezing the GUI.
- **pandas:** This module is used for data manipulation and analysis, particularly for storing and processing attendance data in DataFrames.
- **matplotlib.pyplot:** This module is used for creating visualizations of the attendance data, such as bar charts and pie charts.
- **os:** This module provides functions for interacting with the operating system, such as creating directories and checking file paths.
- **numpy:** This module is used for numerical operations, especially for handling arrays of image data.

6. Technology Selected and Technology features covered:

- **Technology Selected:**
 - Python: The primary programming language used for the entire application.
 - OpenCV: For face detection and recognition.
 - SQLite: For local database management.
 - tkinter: For GUI development.
 - pandas & matplotlib: For data handling and visualization.
- **Technology Features Covered:**

- **Face Detection:** Using Haar cascades to detect faces in real-time video frames.
- **Face Recognition:** Employing the LBPH (Local Binary Patterns Histograms) algorithm for face recognition.
- **Image Capture and Processing:** Capturing images from a webcam, converting them to grayscale, and saving them to a dataset.
- **Model Training:** Training the face recognition model on a dataset of captured face images.
- **Attendance Marking:** Recording the name, date, and time of recognized individuals in a database.
- **GUI Development:** Creating a user-friendly interface with buttons, labels, and message boxes.
- **Data Visualization:** Generating bar charts and pie charts to visualize attendance data.
- **Database Interaction:** Storing and retrieving attendance information using SQLite.
- **Threading:** Using threading to improve application responsiveness during face recognition.

7. References

1. OpenCV Documentation: <https://docs.opencv.org/>
2. Python Official Documentation: <https://docs.python.org/>
3. SQLite Documentation: <https://www.sqlite.org/docs.html>
4. Face Recognition using LBPH Algorithm - Research Papers
5. TutorialsPoint, GeeksforGeeks, and Stack Overflow for code support